COLIN CAMPBELL (CHEMICALS) PTY. LTD.

SAFETY DATA SHEET

Date of Issue: 1 January 2023

1) IDENTIFICATION OF THE MATERIAL AND SUPPLIER

Product Name:	**CAMPBELL VESTA HERBICIDE**
Other Names:	MSMS, monosodium methanearsonate
Chemical Group:	Organo-arsenic compounds
CAS No.:	2163-80-6
Recommended Use:	Herbicide
Supplier Details:	Colin Campbell (Chemicals) Pty Ltd ABN 29 000 045 590 5 Blackfriar Place Wetherill Park NSW 2164
Telephone:	(02) 9725 2544
Fax:	(02) 9604 7768
Email:	cccsyd@campbellchemicals.com.au
Website:	www.campbellchemicals.com.au
Contact:	Product Development Manager – (02) 9725 2544
Emergency Telephone Number:	13 11 26 (Poisons Information Centre)

2) HAZARDS IDENTIFICATION

GHS classification:	Acute toxicity-Oral Category 5 Acute toxicity-Dermal Category 5 Acute-Inhalation Category 4 Hazardous to the aquatic environment Acute hazard Category 3 Long term hazard Category 3
GHS Signal Words:	WARNING
Hazard Statements :	H303 May be harmful if swallowed. H313 May be harmful in contact with skin. H332 Harmful if inhaled. H412 Harmful to aquatic life with long lasting effects.
General Precautionary Statements :	P102 Keep out of reach of children. P103 Read carefully and follow all instructions.

Pictograms :

Precautionary statements Prevention:	P261 Avoid breathing mist. P271 Use only outdoors or in a well-ventilated area. P273 Avoid release to the environment.
Precautionary statements Response:	P312 Call a POISON CENTER/doctor/physician if you feel unwell. P304+P340 IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
Storage :	No specific precautions
Disposal:	P501: Dispose of contents/container in accordance with hazardous waste regulation.
Other information :	Dangerous and hazardous goods as defined by WHS. SUSMP Classification : S7 (DANGEROUS POISON)

3) COMPOSITION/INFORMATION ON INGREDIENTS

Hazardous Ingredients	CAS Number	Concentration
MSMA, Monosodium methanearsonate	2163-80-6	720g/L

4) FIRST AID MEASURES

If poisoning occurs, move out of dangerous area immediately contact a doctor or Poison Information Centre (Ph: 13 11 26) and follow the advice given.
Show this Safety Data Sheet to the doctor.

If inhaled:	Move person to fresh air. If person is not breathing, call 000 or an ambulance, then give artificial respiration, preferably by mouth-to-mouth, if possible. Call a Poison Information Centre or doctor for further treatment advice.
In case of skin contact:	Take off contaminated clothing. Rinse skin immediately with plenty of water for 15-20 minutes. Call a Poison Information Centre or doctor for treatment advice.
In case of eye contact:	Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye. Call a poison control center or doctor for treatment advice.
If swallowed:	Call a Poison Information Centre or doctor immediately for treatment advice. Have person sip a glass of water if able to swallow. Do not induce vomiting unless told to do so by a Poison Information Centre or doctor. Do not give anything by mouth to an unconscious person.
First Aid facilities	Ensure eye wash and safety shower are available.

Medical Attention:

Most important symptoms and effects, both acute and delayed:
Although the organic (methylated) pentavalent compounds (arsonates) incur the least hazard of the arsenicals, it is prudent to manage cases of arsenical pesticide ingestion as though all are highly toxic.

Effects associated with acute MSMA exposure:
Gastrointestinal (GI) adverse effects predominate, with vomiting, abdominal pain and rice-water or bloody diarrhea being the most common. Other GI effects include inflammation, vesicle formation and eventual sloughing of the mucosa in the mouth, pharynx and oesophagus. Symptoms related to the central nervous system may begin with headache, dizziness, drowsiness and confusion.

Effects associated with chronic exposure to MSMA:
Chronic exposure to MSMA is unlikely when handling according to label directions

Indication of any immediate medical attention and special treatment needed.
If ingested, gastric lavage may be indicated. Literature recommendations for arsenic poisoning calls for chelation therapy with BAL or d-penicillamine. Persons with sensitivity to penicillins may suffer an allergic reaction. BAL is recommended for persons allergic to penicillin.

5) FIRE FIGHTING MEASURES

Extinguishing media

Dry chemical or carbon dioxide (CO2), do not use water or foam, as this may spread product.

Hazard from combustion products

Reducing agents may release arsine gas.

Precautions for fighting fires

Fire fighters should wear full protective gear, including self-contained breathing apparatus (AS/NZS 1715/1716). Keep unnecessary people away. If it can be done safely remove intact containers from the fire. Bund area with sand or earth to prevent contamination of drains or waterways. Dispose of fire residues and contaminated fire extinguishing water in accordance with local regulations. Do not release contaminated water into the environment.

Hazchem Code

N/A

6) ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment:

Use personal protective equipment. Avoid breathing mist. Ensure adequate ventilation.

Environmental precautions:

Do not apply directly to water. Do not discharge into drains/surface waters/groundwater. Do not contaminate water by cleaning of equipment or disposal waste. Do not contaminate waters used for domestic purposes, or by

wildlife, including aquatic life, or for irrigation.

Methods and materials for containment and cleaning up: Stop leak if possible. Contain spilled product with an inert dyking material, such as sand. Cover the spill area with a 1:1 mixture of vermiculite and solid calcium oxide (the amount of the mixture should be at least double the size of the spill). Place reclaimed product in a closed and properly labelled waste drum. Store drum in separate area until proper disposal. Flush residue with water.

7) HANDLING AND STORAGE

Handling Wear suitable protective clothing. Avoid contact with skin and eyes. Do not breathe mist. Avoid formation of aerosols. Provide appropriate exhaust ventilation. Wash hands and exposed skin thoroughly after handling.

Storage Store in a cool place. Keep container tightly closed in a dry and well ventilated place. Damaged or leaking containers which cannot be used immediately should be transferred to suitable sound containers and properly marked.
Incompatibility: Heavy metal salts may cause precipitation. Mild corrosive to steel. May react with reducing agents such as aluminum, zinc, sodium borohydride, sulfur dioxide.

8) EXPOSURE CONTROL/PERSONAL PROTECTION

Exposure Standards

Exposure limits for organic arsenic compounds
OSHA PEL-TWA, (as As):
0.5 mg/m^3
OSHA California/PEL-TWA, (as As):
0.2 mg/m^3

TWA	TLV	STEL
Not established	Not established	Not established

Exposure standard – **Time Weighted Average (TWA)** means the average airborne concentration of a particular substance when calculated over a normal eight hour working day, for a five-day working week.

Short Term Exposure Limit (STEL) means the exposure level that may be equalled (but should not be exceeded for no longer than 15 minutes and should not be repeated more than 4 times per day. There should be at least 60 minutes between successive exposures at the STEL.

Biological Limit Values None allocated

Engineering Controls Provide exhaust ventilation or other engineering controls.
Wash hands before breaks and at the end of the workday.

Personal Protective Equipment

Eyes:	Safety goggles. Eye wash bottle with pure water.
Clothing:	Long sleeve shirt, long pants, boots.
Gloves:	Chemical resistant gloves. Before removing gloves clean them with soap and water.

Respiratory: Dust or mist respirator.

9) PHYSICAL AND CHEMICALS PROPERTIES

Appearance:	Clear colourless to yellow liquid.
Odour:	Odourless
pH:	5-7
Vapour pressure:	<1x10-5 Pa (MAA) @25°C
Vapour density:	Not available
Boiling point:	Not available
Freezing point:	-25 to -15°C
Solubility in water (25°C):	104 g/100 mL (MSMA 100%)
Bulk density (20°C):	1.3-1.7 g/mL
Viscosity (25°C):	27.7 cps (MSMA 51%)
Flash point:	>100°C
Flammability (explosive) limit:	Not flammable
Auto ignition temperature:	Not available
Partition coefficient (octanol/water):	Not available

10) STABILITY AND REACTIVITY

Chemical stability:	Stable under normal temperatures and pressure.
Conditions to avoid:	Not available.
Incompatible materials :	Reducing agents such as aluminium, zinc, sodium borohydride, sulphur dioxide.
Hazardous decomposition products:	Reducing agents may release arsine gas.
Hazardous reactions:	Not available.

11) TOXICOLOGICAL INFORMATION

Inhalation:	No data available.
Skin contact:	Causes mild skin irritation.
Eye contact:	Causes eye irritation.
Ingestion:	No data available.
Chronic toxicity:	
Germ cell mutagenicity:	Negative
Carcinogenicity:	Negative

Toxic to reproduction:	Negative.
Specific target organ toxicity (repeated exposure):	No known target organ effects. May cause redness and reversible irritation to eyes. Prolonged or repeated contact may cause skin irritation.
Acute toxicity:	
Oral toxicity:	Oral LD50 (rat): 2833 mg/kg
Dermal toxicity :	Dermal LD50 (rabbit): >2000 mg/kg
Inhalation toxicity:	Inhalation LC50 (1h, rat): 10.8 mg/L Inhalation LC50 (4h, rat): 2.2 mg/L
Skin corrosion/irritation: Skin irritation (rabbit):	Not a primary irritant
Sensitisation:	Dermal sensitization (guinea pig): Not a contact sensitizer
Respiratory sensitisation	No information
Specific target organ toxicity (STOT) (single exposure):	**STOT single exposure:** Not available **STOT repeated exposure:** Not available
Germ cell mutagenicity:	Not mutagenic
Carcinogenicity:	MSMA is not a carcinogen. It did not cause tumors in any of the carcinogenicity studies that were conducted.
Reproductive toxicity:	Not available
Aspiration hazard:	Not available

12) ECOLOGICAL INFORMATION

DO NOT contaminate streams, rivers or waterway with this product or the used containers.

Ecotoxicity:	**Ecotoxicity: (MSMA 51%)** **Birds:** Northern bobwhite LD50: 834 mg/kg Northern bobwhite LC50 (5d): 3269 mg/L Mallard LC50 (5d): >5620 mg/L **Fish:** Bluegill sunfish LC50 (96 h): > 93 mg/L **Crustacea:** *Daphnia pulex* LC50 (48 h): 82.85 mg/L **Aquatic plants:** *Lemna gibba* ErC50 (14 days): 145.9 mg /L **Bees:** *Apis mellifera* LD50 (48hr, contact): 68 □g/bee
Environmental fate, persistence and degradability, mobility	Not readily biodegradable

Bioaccumulative potential:	No bio-accumulative potential (the product is hydrophilic, BCF les than 1x
Other precautions:	Do not contaminate dams, waterways or sewers with this product or used containers.

13) DISPOSAL CONSIDERATIONS

This product should not be allowed to enter drains, water courses or the soil. Do not contaminate ponds, waterways or ditches with chemical or used containers. Triple or preferable pressure rinse containers before disposal. Add rinsings to the mixing tank. Do not dispose of undiluted chemical on-site. If recycling, replace cap and return clean containers to recycler or designated collection point. If not recycling, break, crush or puncture and bury empty containers in a local authority landfill. If no landfill is available bury the containers below 500mm in a disposal pit specifically marked and set up for this purpose clear of waterways, desirable vegetation and tree roots. Empty containers and product should not be burnt. Do not reuse empty containers.

14) TRANSPORT INFORMATION

International regulation

UNRTDG	Not regulated as a dangerous good.
IATA-DGR	Not regulated as a dangerous good
IMDG-Code	Not regulated as a dangerous good.

National Regulations

ADG Code	Not considered as a dangerous good for transport by road or rail in Australia under the Australian Dangerous Goods Code.

15) REGULATORY INFORMATION

This data sheet complies with the requirements of the Global Harmonized System of Classification and Labelling of chemicals (GHS).

16) OTHER INFORMATION

Date of revision : 1 January 2023
Reason for revision : New product.

This SDS summarises our best knowledge of the health and safety hazard information of the product and how to safely handle and use the product in the workplace. Each user should read this SDS and consider the information in the context of the how the product will be handled and used in the workplace including in conjunction with other products.

END OF SDS